

Physical W -shape for the ϕ diffractive generator: fixing the near-threshold cross section from photoproduction

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Abstract

The tuned ϕ electroproduction cross section used a threshold factor $(1 - W_{th}^2/W^2)^{\alpha_2}$ with a *negative* exponent ($\alpha_2 = -1.245$), which diverges at threshold – unphysical, and it blows up the radiative-correction (RC) integral. We show that CLAS12 data cannot constrain the near-threshold W -shape, fix that shape to real-photon photoproduction (Roberts et al., Eq. 23b: $\alpha_2 = 2$, $\alpha_3 = \epsilon = 0.20$), and refit only the two parameters CLAS12 does constrain – the t -slope b_t and the Q^2 exponent ν_T . The result describes the data equally well ($\chi^2 = 13.0$ vs 11.3), is physical at threshold, and makes the RC well-behaved (bounded η , no blow-ups).

1 The problem

The ϕ generator models the transverse cross section as

$$\frac{d\sigma_T}{dt} = \alpha_1 \left(1 - \frac{W_{th}^2}{W^2}\right)^{\alpha_2} W^{\alpha_3} \left(1 + \frac{Q^2}{m_\phi^2}\right)^{-\nu_T} b_t e^{b_t t}, \quad \frac{d\sigma_L}{dt} = c_R \frac{Q^2}{m_\phi^2} \frac{d\sigma_T}{dt}, \quad (1)$$

with $W_{th} = m_p + m_\phi = 1.96$ GeV. The original tuning returned $\alpha_2 = -1.245$. A *negative* threshold exponent makes $(1 - W_{th}^2/W^2)^{\alpha_2} \rightarrow \infty$ as $W \rightarrow W_{th}$: the cross section *diverges* at threshold instead of vanishing (Fig. 1). This is unphysical, and it is fatal for the radiative correction: the RC integral samples σ_T at the radiatively-shifted $W' = \sqrt{W^2 - v} < W$, which is dragged toward W_{th} , so the integrand explodes. In an independent RC scan (60-point grid) the old model gave $\eta = \sigma_{obs}/\sigma_{Born}$ up to 13 with MC variance as large as the value, at $\sim 1/3$ of grid points.

2 Why CLAS12 cannot fix the W -shape

The CLAS12 ϕ data span $W \in [2.0, 3.5]$ GeV – the lower edge is only ~ 40 MeV above threshold, and W is strongly correlated with Q^2 in fixed-target electroproduction. Consequently the W -shape is essentially unconstrained: fitting α_2, α_3 freely gives per-dataset α_3 scattered over 0.22–1.44 (a factor 6), and the fit exploits the *negative* α_2 merely as an extra in-range shape knob with no physical meaning. Forcing $\alpha_2 \geq 0$ while still *fitting* α_3 does not help: α_2 pins to 0, α_3 swings to ≈ -3 , and χ^2 triples ($11 \rightarrow 32$). The data simply do not know the W -shape.

3 The fix: take the W -shape from photoproduction

Real-photon photoproduction *does* measure the W -shape, over a wide clean range. Roberts et al. (arXiv:2510.08845) parametrise the total cross section with the all- W Donnachie–Landshoff form

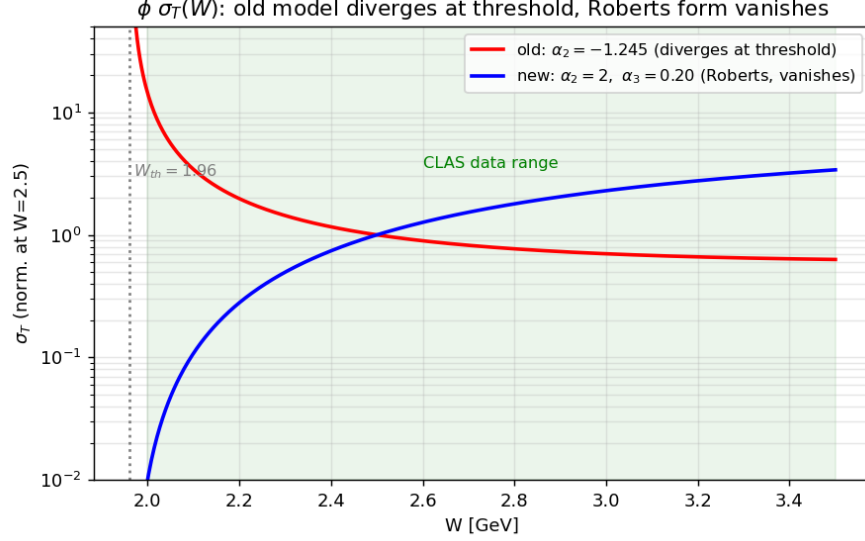


Figure 1: $\phi \sigma_T(W)$ near threshold (log scale, normalised at $W = 2.5$). The old model ($\alpha_2 = -1.245$, red) diverges as $W \rightarrow W_{th}$; the Roberts form ($\alpha_2 = 2$, $\alpha_3 = 0.20$, blue) vanishes at threshold and rises with W as the Pomeron dictates. The CLAS data live in the shaded band, ~ 40 MeV above threshold.

(their Eq. 23b),

$$\sigma(W) = \sigma_{\mathbb{P}} \left[1 - \frac{(m_p + m_V)^2}{W^2} \right]^2 W^\epsilon, \quad (2)$$

which is *identical in structure* to the W -part of Eq. (1). For ϕ their P-dyn fit gives $\epsilon = 0.20$. We therefore fix

$$\boxed{\alpha_2 = 2 \quad (\text{threshold, } \sigma_T \rightarrow 0), \quad \alpha_3 = \epsilon = 0.20 \quad (\text{Roberts P-dyn } \phi)} \quad (3)$$

and fit only b_t and ν_T to CLAS12. The normalisation α_1 stays a free scale (the tuning fits shapes, not absolute normalisation); $c_R = 1$ fixed. The exponent 2 is the DL convention – what physics requires is only that it be positive (phase space gives $\geq 1/2$); the value does not affect the CLAS fit because b_t, ν_T are orthogonal to the W -shape.

4 The fit

For each of the five CLAS12 RGA datasets we generate ϕ MC with the fixed W -shape, apply the neural-network fast simulation for acceptance, extract the ϕ signal by fitting $M(K^+K^-)$ in each kinematic bin (Fig. 2), acceptance-correct the data, and fit b_t, ν_T to the corrected W, Q^2, t distributions (Figs. 3, 5). Only b_t and ν_T are free.

The result (Table 1) has $\chi^2 = 13.0$, statistically identical to the old free fit (11.3), with b_t, ν_T essentially unchanged and now *tighter*. The negative- α_3 pathology is gone: because we no longer ask the data to set the W -shape, there is nothing to force α_3 negative.

Note on iteration. The standard tuning iterates generate $\rightarrow$ refit. With the W -shape fixed, that loop does *not* converge: χ^2 stays flat at 13.0 while b_t, ν_T drift and the inter-dataset spread grows (an

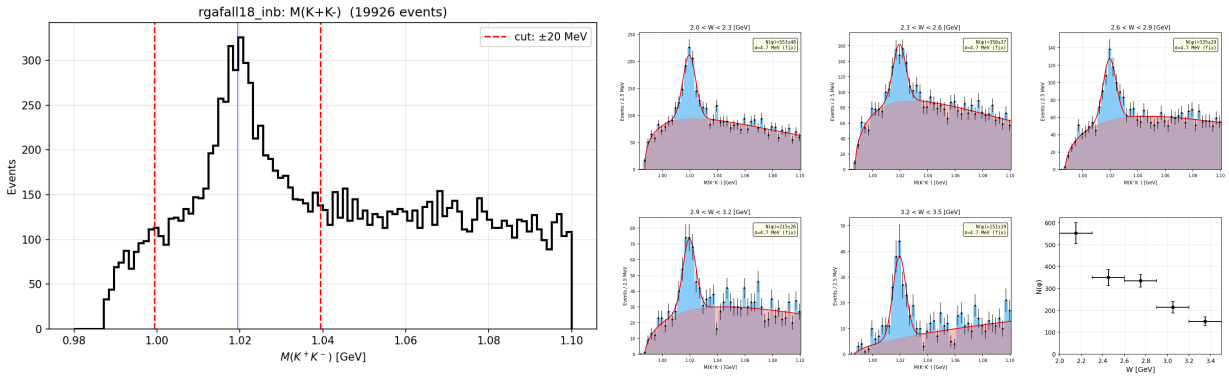


Figure 2: Signal extraction (Fall18-inb). Left: inclusive $M(K^+K^-)$ with the ϕ peak. Right: the ϕ yield from fitting $M(K^+K^-)$ in bins of W .

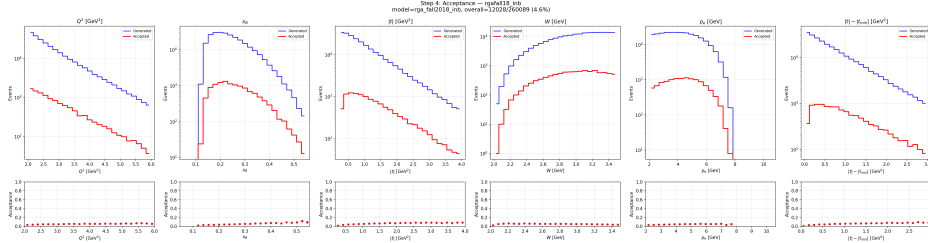


Figure 3: Acceptance from the NN fast simulation vs. W , Q^2 , t , x_B (Fall18-inb).

unstable regenerate-refit map along a flat $b_t - \nu_T$ valley). A single reweighting pass already recovers the answer; iteration is unnecessary and destabilising here. The single-pass values in Table 1 are the result.

5 Radiative-correction verification

Re-running the RC 60-point grid with the new W -shape confirms the fix: η is bounded to 0.745–0.840 with **0/60 blow-ups** (Fig. 6), versus 19/60 diverging points for the old model. The model dependence of the RC is small ($\sim 6\%$ in the diffractive region; Fig. 6, right).

6 Where the parameters live (three directories)

The final ϕ parameters are used in three separate code bases. They must be kept consistent; if the ϕ tuning is ever redone, *all three* must be updated together.

All three now carry $\alpha_2 = 2$, $\alpha_3 = 0.20$, $\nu_T = 2.253$, $b_t = 1.326$, $c_R = 1$. In `diffrad.vm.f90` and the tuning generator these are the compiled defaults (overridable by input-file keys `alf1 alf2 alf3 nuT bt cR`); in `diff1t` they are hard-coded in the `ivec=3` branch.

7 Summary

- The old ϕ W -shape ($\alpha_2 = -1.245$) was unphysical (diverges at threshold) and broke the RC.

Table 1: Final ϕ fit. $\alpha_2 = 2$, $\alpha_3 = 0.20$ fixed (Roberts Eq. 23b, P-dyn); $c_R = 1$; α_1 free. Only b_t, ν_T fit.

dataset	b_t [GeV $^{-2}$]	ν_T	χ^2
Fall18 inb	1.248	2.327	14.9
Fall18 outb	1.470	2.557	18.4
Sp18 inb	1.348	2.072	9.6
Sp18 outb	1.321	2.237	11.8
Sp19 inb	1.245	2.074	10.2
average	1.326$\pm$0.082	2.253$\pm$0.181	13.0

Table 2: The final ϕ cross-section parameters and where they are set. Averaged values: $\alpha_2 = 2$, $\alpha_3 = 0.20$, $b_t = 1.326$, $\nu_T = 2.253$, $c_R = 1$, α_1 free.

Directory	File (function)	Role
vector_mesons_generator_tuning	phi/diffrad_vpk_exp.f90 + phi/runs/*/config.json	tuning generator (determines params)
MC_vector_mesons	diffrad_vm.f90 (sigma_T_phi)	production / RC generator
RC_vector_mesons_DIFFRAD	tuned_roberts_phi_run/mdiffrad.f (diff1t)	RC η -grid calculation

- CLAS12 cannot determine the near-threshold W -shape.
- Fixing it to photoproduction (Roberts Eq. 23b P-dyn: $\alpha_2 = 2$, $\alpha_3 = 0.20$) and fitting only $b_t = 1.326 \pm 0.082$, $\nu_T = 2.253 \pm 0.181$ describes the data equally well ($\chi^2 = 13.0$), is physical at threshold, and makes the RC bounded.
- Free parameters: b_t, ν_T (fit); α_1 (free normalisation). Fixed: α_2, α_3 (photoproduction), c_R .

phi fit (Roberts Eq.23b: $\alpha f_2=2$, $\alpha f_3=0.20$ fixed; bt, nuT free) -- data vs best-fit model, 5 datasets x 4 variables

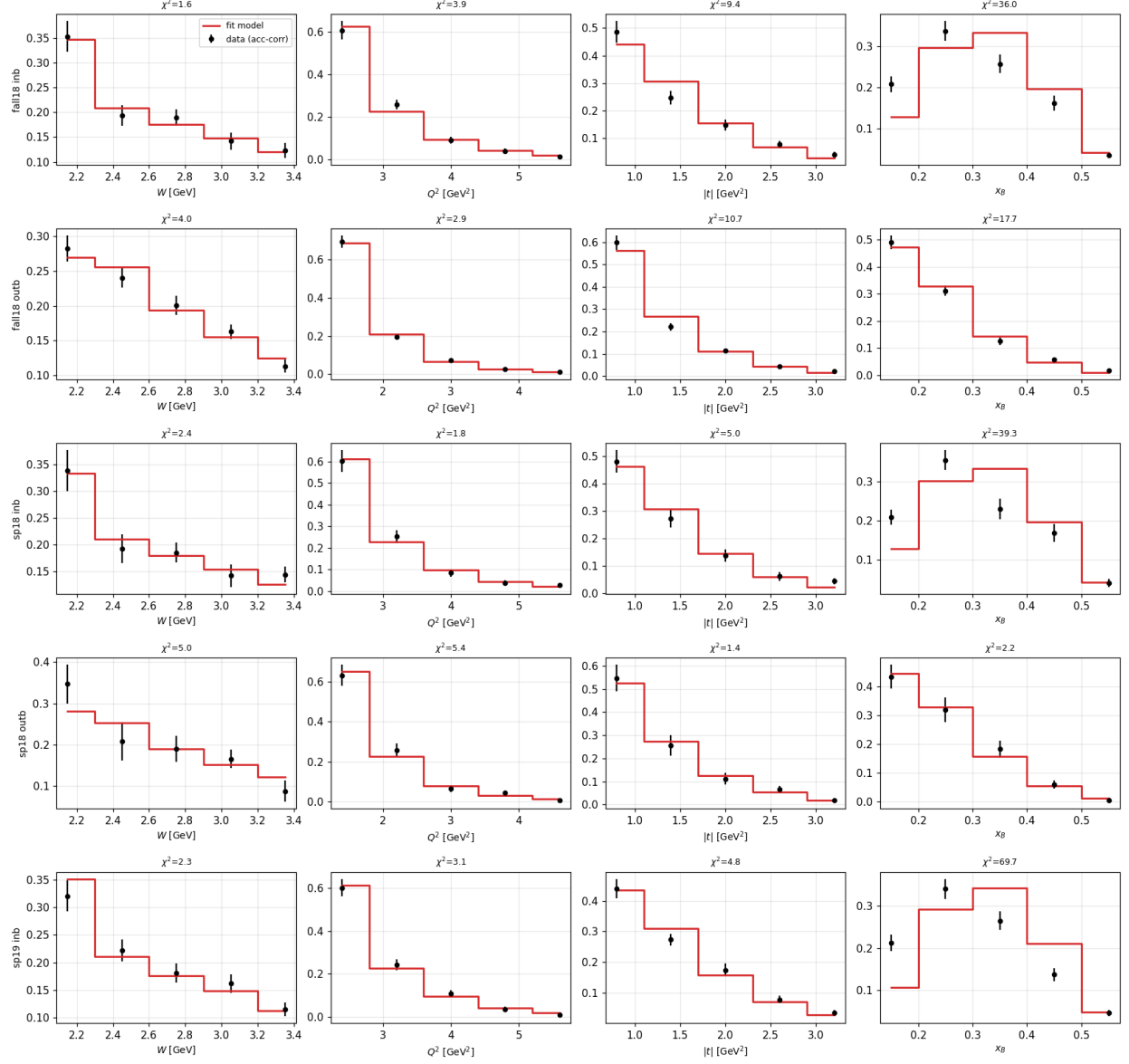


Figure 4: Acceptance-corrected data (black) vs. best-fit model (red) for all five datasets $\times$ four variables ($W, Q^2, |t|, x_B$). χ^2 per panel shown; the fit uses W, Q^2, t .

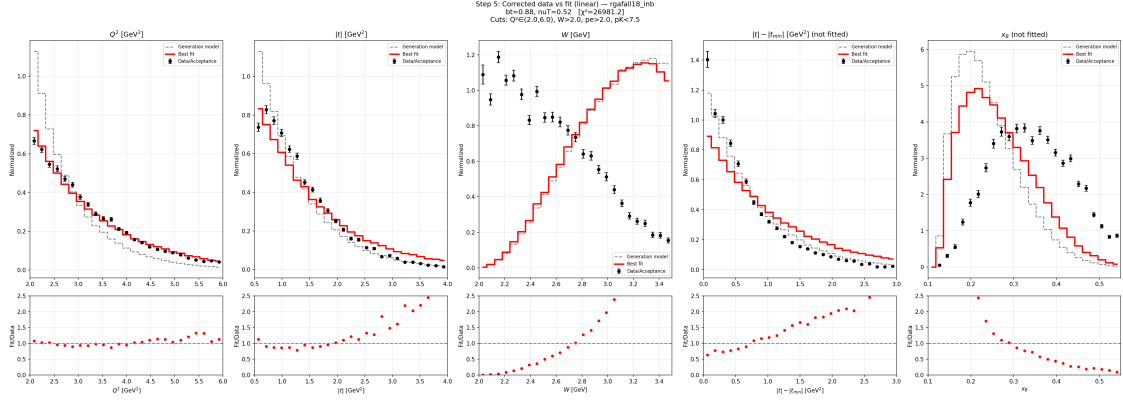


Figure 5: Data vs. best-fit model, Fall18-inb (detail).

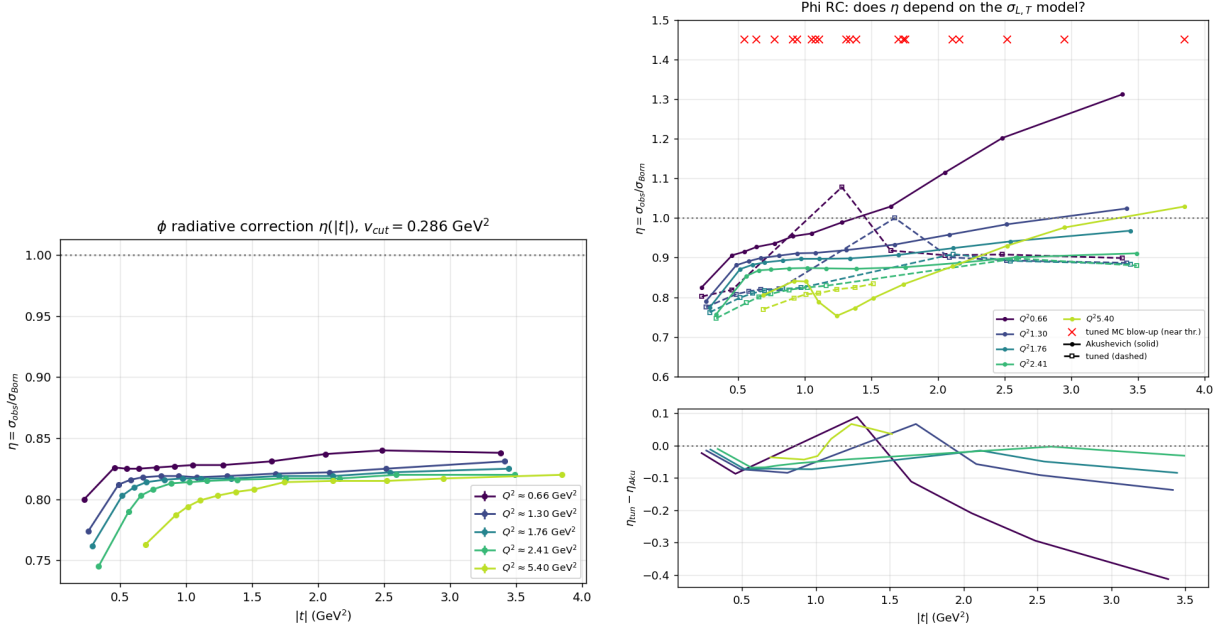


Figure 6: Left: RC factor $\eta(|t|)$ with the Roberts W -shape – smooth, bounded, physical. Right: η model comparison (Akushevich vs. tuned), $\sim 6\%$ in the diffractive region.